

ALEXANDER IMMANUEL GOERKE ZAMORA

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SUMMARY

Industrial Electronics and Automation Engineer from Universitat Politècnica de València, with a year at the Technical University of Denmark focused on robotics and digital control. Experienced across the full engineering cycle, from theoretical derivation and simulation to real hardware implementation, experimental validation, and deployment in industrial environments, within international and multidisciplinary teams.

EDUCATION

IALE INTERNATIONAL SCHOOL

Secondary Education (ESO & Bachillerato)

- Awarded the Science Prize for academic excellence.

UNIVERSITAT POLITÈCNICA DE VALÈNCIA

B.Sc. Industrial Electronics and Automation Engineering

TECHNICAL UNIVERSITY OF DENMARK

Erasmus+ Exchange

Valencia, Spain

2016 – 2022

Valencia, Spain

2022 – 2026

Lyngby, Denmark

2025 – 2026

EXPERIENCE

CEBREO MEDICAL A/S

Software Engineer

- Developed signal-processing algorithms and visualization tools for biomedical signals (EEG, PPG, and triaxial accelerometer data), including filtering, pulse-analysis, and posture-detection algorithms based on accelerometer signals, supporting clinical biosignal inspection across EDF, CSV, and .mat data formats.
- Designed synthetic test-signal pipelines for algorithm validation and QA, working within a multidisciplinary team of software engineers, physiologists, and product developers.

Copenhagen, Denmark

Jan 2026 – Jun 2026

PRIVATE TUTORING

Maths, Physics & Chemistry Tutor

- Provide one-to-one tutoring in mathematics, physics, and chemistry for secondary school students since 2022, planning and delivering personalized study plans and applying structured problem-solving to improve academic outcomes.
- Recommended by school teachers for consistently strong academic results, with a focus on clear communication and mentoring.

Valencia, Spain

2022 – Present

PROJECTS

BACHELOR THESIS: DELTA ROBOT

Technical University of Denmark, 2025–2026

- Designed the control system for a delta robot supporting a 5-year PhD programme on agricultural weed-intervention, deriving full IK/FK kinematics, building a complete simulation suite in Python, and porting the controller to C++ deployed on a Teensy 4.1.
- Validated feedforward, smooth bounded feedback, and full PID strategies, achieving 1–2 mm end-effector positioning accuracy through quantitative real-robot experiments.

DIFFERENTIAL-DRIVE ROBOT

Technical University of Denmark, 2025

- Ranked 1st in class. Designed PI/PD controllers with encoder-based sensor fusion, validated in Simulink and deployed on a Teensy 4.1.

VISION-GUIDED 4-DOF ROBOT

Technical University of Denmark, 2025

- Developed the vision-to-motion pipeline for a 4-DOF robotic arm, converting pixel coordinates to world space and reaching targets via inverse kinematics, with HSV segmentation, camera calibration, and Dynamixel servo control.

TECHNICAL SKILLS

- Engineering:** Robotics, Digital Control, Signal Processing, Embedded Systems, Circuit Design, IK/FK Kinematics, Sensor Fusion, Mechatronics
- Programming:** Python, C/C++, MATLAB, Arduino, HTML/CSS
- Design & Simulation:** SolidWorks, AutoCAD, MATLAB/Simulink, Proteus, OrCAD
- Tools & AI:** Git, VSCode, LaTeX, Claude, Codex

CERTIFICATIONS & AWARDS

- ISO 21500 & ISO 21502: Project, Programme and Portfolio Management (Danish Standards, Jan 2026)
 - Advanced English (C1) | German (A2) | Web Programming (HTML/CSS/JS) | Energy Systems (CFP, 2024)
- Languages:** Spanish (native) | English (C1) | German (A2)